

I. Confidence Intervals and Hypothesis Testing (Lec 10)

1. **Answer:** 0.01
Explanation: The significance level $\alpha = 1 - \text{Confidence Level} = 1 - 0.99 = 0.01$.
2. **Answer:** A range of values that is likely to contain the population parameter with a given confidence.
Explanation: CI represents uncertainty around a sample estimate.
3. **Answer:** Population assumptions or claims.
Explanation: Hypothesis testing validates whether a population assumption holds true using sample data.
4. **Answer:** $(Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}})$
Explanation: This Z-statistic tests how far a sample mean is from the population mean.
5. **Answer:** When the population standard deviation (σ) is unknown and sample size is small.
Explanation: The t-test compensates for extra uncertainty in small samples.
6. **Answer:** $(E[\hat{\beta}] = \beta)$
Explanation: OLS estimators are unbiased under classical linear model assumptions.

II. Overfitting, Underfitting, and Ridge Regression (Lec 11)

7. **Answer:** Overfitting.
Explanation: The model memorizes training data and performs poorly on unseen test data.
8. **Answer:** High loss on both training and test sets.
Explanation: Underfitting means the model is too simple to capture the pattern.
9. **Answer:** Training data noise.
Explanation: Overfitting models capture random fluctuations in training data.
10. **Answer:** To reduce model complexity and prevent overfitting.
Explanation: The penalty term shrinks coefficient magnitudes.

11. **Answer:** Regularization parameter (λ).
Explanation: λ controls the strength of the penalty on coefficients.

12. **Answer:** When $(X^T X)$ is not invertible (singular).
Explanation: Ridge adds λI to make the matrix invertible.

III. LASSO and Elastic Net Regression (Lec 12)

13. **Answer:** Least Absolute Shrinkage and Selection Operator.
Explanation: LASSO performs both shrinkage and feature selection.

14. **Answer:** Some coefficients become exactly zero in LASSO.
Explanation: LASSO produces sparse solutions by enforcing L1 regularization.

15. **Answer:** Absolute value function ($|\beta|$).
Explanation: LASSO uses L1 penalty, summing absolute coefficient magnitudes.

16. **Answer:** Because L1 regularization is non-differentiable at zero.
Explanation: Analytical derivation becomes impossible.

17. **Answer:** Numerical optimization methods (e.g., coordinate descent).
Explanation: Used to iteratively minimize the LASSO loss.

18. **Answer:** Ridge (L2) and LASSO (L1) regression.
Explanation: Elastic Net combines both penalties for better stability.

IV. Bias-Variance Tradeoff (Lec 13)

19. **Answer:** The relationship between model complexity and prediction error.
Explanation: Helps understand why models underperform or overfit.

20. **Answer:** The error from overly simplistic assumptions.
Explanation: Bias causes systematic deviation from true values.

21. **Answer:** The sensitivity of the model to small fluctuations in the training set.
Explanation: High variance means model performance changes a lot between datasets.

22. **Answer:** Underfitting.

Explanation: High bias models fail to learn the underlying trend.

23. **Answer:** $E[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$

Explanation: Decomposition of prediction error.

24. **Answer:** Increase model complexity.

Explanation: More parameters help capture relationships better.

25. **Answer:** Use ensemble methods (e.g., bagging).

Explanation: Averaging reduces variance between models.

V. Logistic Regression and Classification Metrics (Lec 14)

26. **Answer:** Binary classification problems.

Explanation: Logistic regression models probabilities for two outcomes.

27. **Answer:** 0.5

Explanation: Predictions ≥ 0.5 are classified as positive (1).

28. **Answer:** Actual = 1, Predicted = 0

Explanation: False Negative means the model missed a positive case.

29. **Answer:** $\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$

Explanation: Measures overall correct predictions.

30. **Answer:** Because it ignores class imbalance and may favor majority class.

Explanation: Other metrics like F1-score are better for imbalanced data.